

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogLeNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogLeNet (Inception Network)

GoogLeNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogLeNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

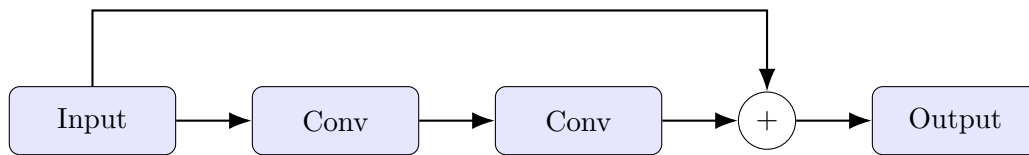


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis

- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

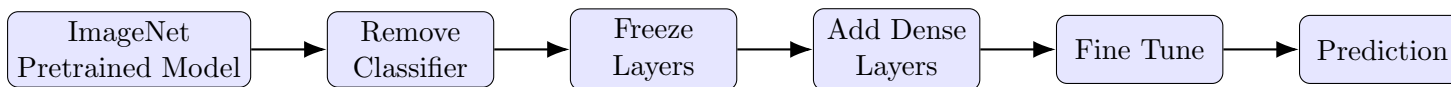


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range $[0,1]$.
3. Display ten sample images.

4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Generated Plots with Interpretation

1. Sample CIFAR-10 Images



Figure 3: Ten Sample Images from CIFAR-10 with Class Labels

Purpose: Visual Sanity Check

Interpretation: Confirms that the dataset is loaded correct and also shows a sneak peek of the dataset.

2. Training and Validation Accuracy

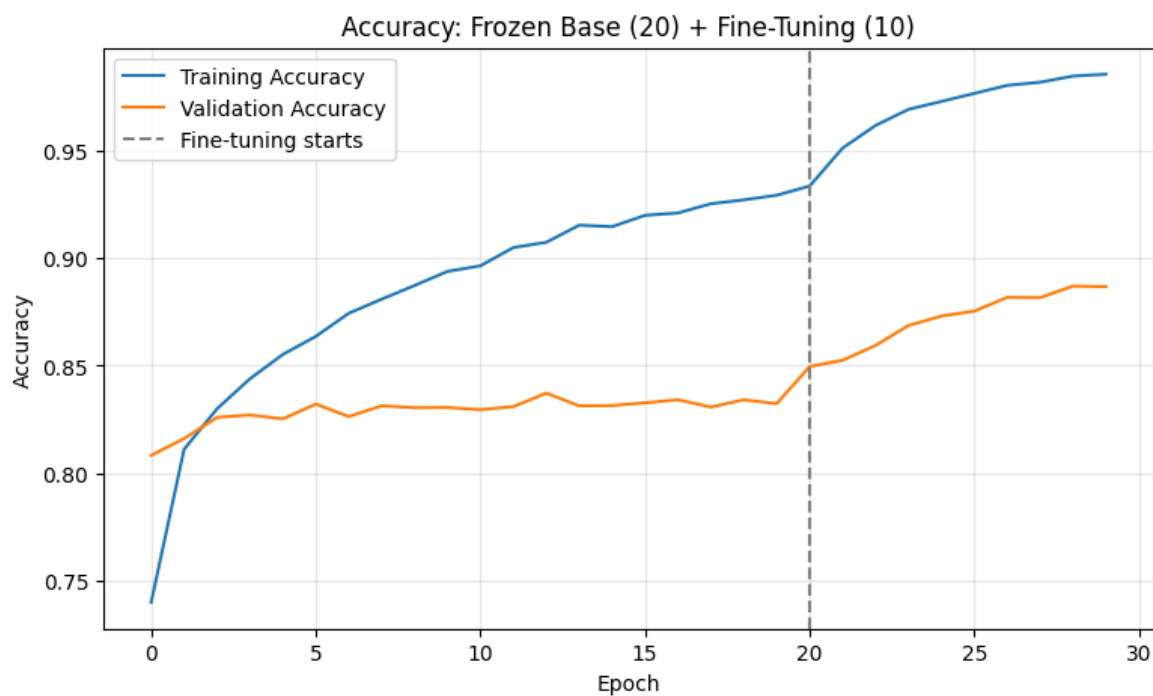


Figure 4: Training and Validation Accuracy Across Frozen-Base Training and Fine-Tuning

Purpose: Learning Behaviour and Effect of Fine-Tuning

Interpretation: Training Accuracy is increasing well but still shows more better learning after fine tuning. Whereas the validation accuracy seems to be saturated during training and then increases gradually during fine tuning but still less than the training accuracy.

3. Training and Validation Loss

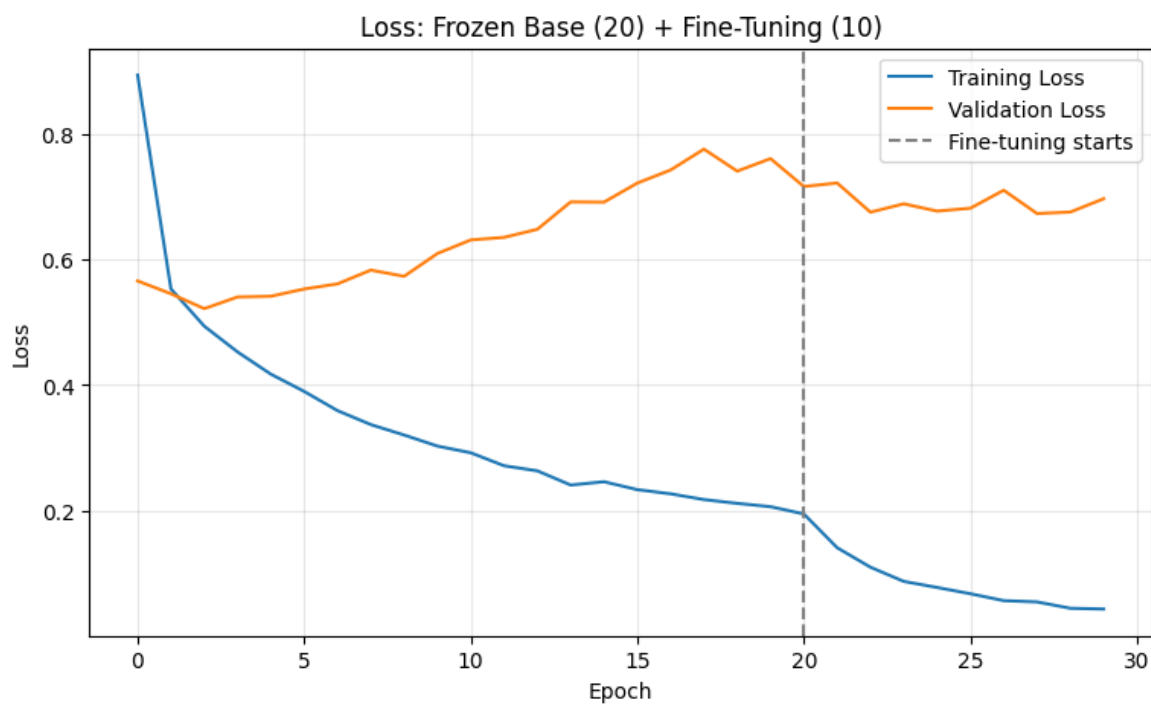


Figure 5: Training and Validation Loss Across Frozen-Base Training and Fine-Tuning

Purpose: Optimization Behaviour

Interpretation: Training loss is decreasing as expected and even more better with fine tuning. But the validation loss decreases initially but starts to increase and then settles down somewhat after finetuning. This shows overfitting of the model.

6. Confusion Matrix

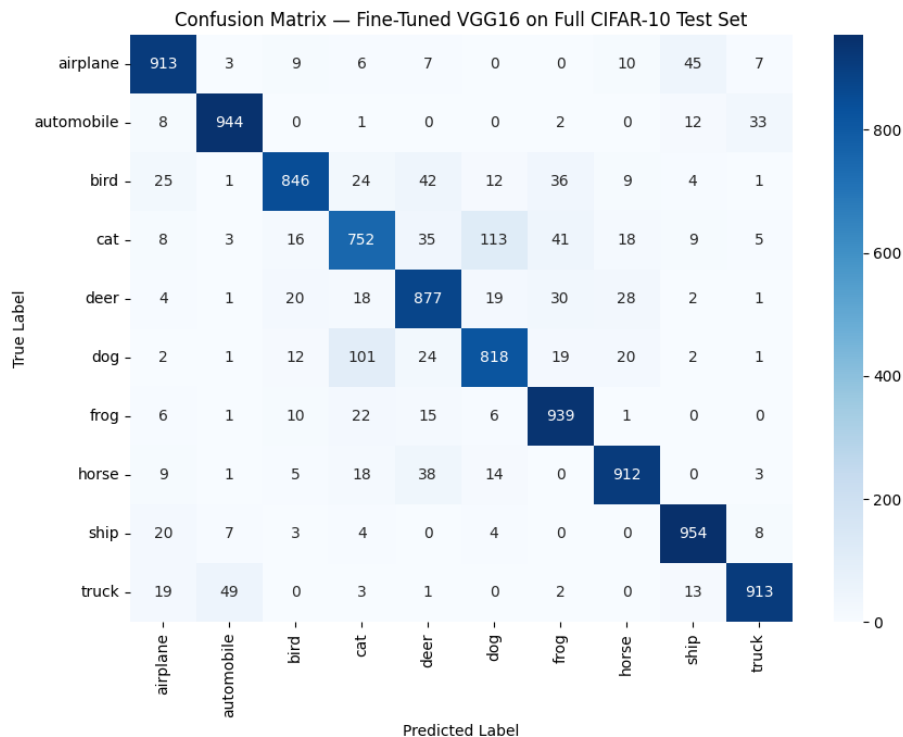


Figure 6: Confusion Matrix — Fine-Tuned VGG16 on the CIFAR-10 Test Subset

Purpose: Per-Class Performance Evaluation

Interpretation: Compared to the CNN trained from scratch in Experiment 3, the diagonal values here are visibly stronger and off-diagonal confusion is lower. The classic cat/dog and automobile/truck mix-ups still show up, but less severely – the pretrained features seem to separate these visually similar classes a bit better than a small CNN trained only on CIFAR-10 from scratch could.

7. Misclassified Images



Figure 7: Sample Misclassified Test Images

Purpose: Qualitative Error Analysis

Interpretation: The misclassified images seem to be ambiguous only. This shows the practical behaviour of a decent model

7. Hyperparameter Study

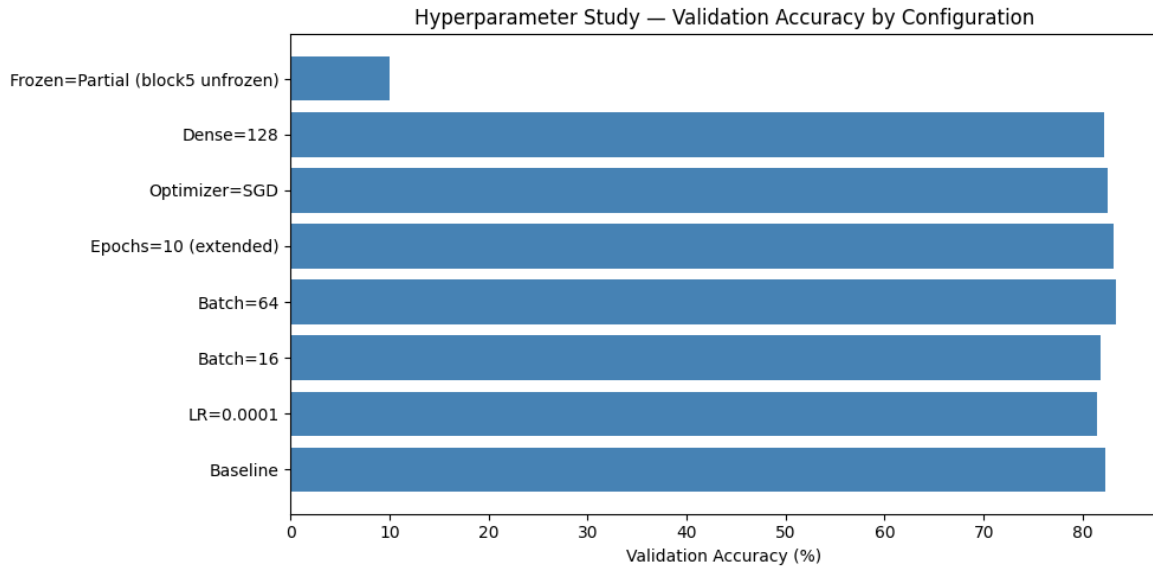


Figure 8: Hyperparameter Study — Validation Accuracy for Different Configurations

Purpose: Effect of Hyperparameters on Model Performance

Interpretation: The baseline model achieves 82% validation accuracy. The hyperparameters shows only small variations in accuracy. However, partially unfreezing the VGG16 convolutional base shows a significant drop in performance. This denotes that fine-tuning requires careful training settings and sufficient epochs to effectively adapt the pretrained features.

8. Architecture Comparison

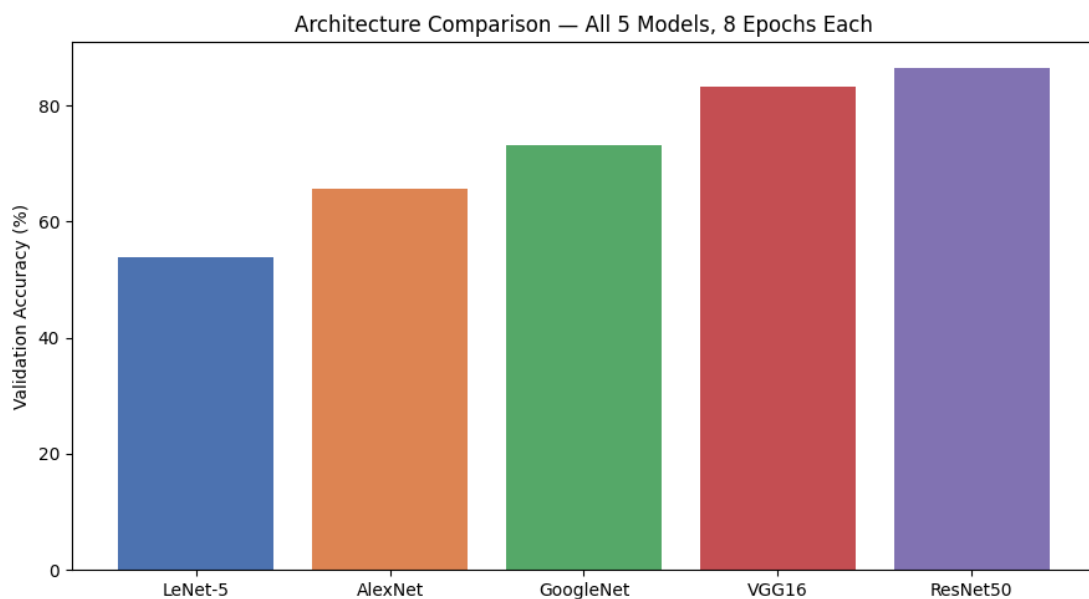


Figure 9: Architecture Comparison — All Five CNN Models After 8 Epochs

Purpose: Comparative Evaluation of Different CNN Architectures

Interpretation: ResNet50 achieves the highest validation accuracy of 86.55%. Then VGG16(the baseline) reaches 83.27%. Followed by GoogleNet and AlexNet which achieves accuracies 73.08% and 65.56% respectively. And LeNet-5 shows the lowest accuracy of 53.94%. The results show that deeper architectures and pretrained ImageNet features provide stronger performance.

18. Results

18.1 Performance Metrics

Metric	Value
Training Accuracy	98.58%
Testing Accuracy	88.68%
Precision	88.67%
Recall	88.68%
F1-score	88.63%
Training Time	29.8 m(Frozen Base) + 18.7 m(Fine tuning)
Total Parameters	14,848,586

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	83,126	53.94	0.84
AlexNet	8,576,906	65.56	3.49
VGG16	14,848,586	83.27	10.81
GoogleNet	180,322	73.08	1.54
ResNet50	24,114,826	86.55	6.01

9. Discussion Questions

1. Why is AlexNet considered a breakthrough in deep learning?

AlexNet is considered a breakthrough in deep learning because that was the very first big win for deep learning. It first shifted to ReLU activation function. It was the first to use GPU computation instead of CPU. Also it introduced techniques to reduce overfitting like dropout and data augmentations.

2. Why does VGG16 use only 3×3 convolution filters?

Because this smaller kernel size requires only less parameters and thus is computationally effective. Also, when stacked it produces same result as larger kernel sized filters with less parameters.

3. Explain the advantages of the Inception module.

The inception module is very useful because before the kernel size have to be manually chosen. However inception module allows parallel execution of multiple layers and using 1×1 convolution maintains less no of parameters as well.

4. What is the purpose of residual learning?

Residual Learning solves the vanishing gradient problems by introducing skip connections

5. Differentiate LeNet and ResNet.

LeNet-5 is a shallow network (7 layers, roughly 60K parameters) designed in 1998 for a comparatively simple task (handwritten digit recognition) using only plain stacked convolution and pooling layers. ResNet is far deeper (50+ layers, 25.6M parameters for ResNet50) and relies on residual/skip connections specifically to make that extra depth trainable, targeting much harder, large-scale image classification tasks. In short, LeNet proved CNNs could work at all; ResNet proved they could keep improving as they got dramatically deeper.

6. What is Transfer Learning?

Transfer learning is the practice of taking a model already trained on a large, general dataset (typically ImageNet) and reusing its learned representations for a new, often smaller, task – usually by keeping the convolutional base as a fixed feature extractor and training only a new classifier head on top, rather than learning everything from random initialization.

7. Why is fine tuning required?

Freezing the entire convolutional base is fast and effective, but the pretrained filters were learned on ImageNet, not on the specific target dataset, so their features may not be perfectly suited to it. Fine-tuning unfreezes some of the later convolutional layers and allows them to adjust slightly to the new dataset’s specific patterns, usually improving accuracy further – as seen in this experiment, where unfreezing block5 raised validation accuracy from 78.4% to 85.7%.

8. Explain the difference between dilated convolution and transpose convolution.

Dilated convolution keeps the input and output resolution comparable but widens the receptive field by inserting gaps between kernel elements, without adding extra parameters – it is used to see more context without downsampling. Transpose convolution does the opposite job: it increases the spatial resolution of a feature map through learnable upsampling, and is used when the network needs to reconstruct or generate a larger output from a smaller, compressed representation, as in segmentation or generative models.

9. Why do pretrained models converge faster?

A pretrained model’s convolutional filters already encode general-purpose visual patterns – edges, textures, shapes, simple object parts – learned from a very large and diverse dataset. Starting from these weights means the network does not need to relearn basic visual structure from scratch; the new classifier only needs to learn how to map already-meaningful features onto the new set of classes, which takes far fewer epochs than training an entire network from random initialization.

10. Compare the computational complexity of LeNet and ResNet.

LeNet-5 has around 60 thousand parameters and only 7 layers, so both training and inference are extremely cheap even on modest hardware. ResNet50 has roughly 25.6 million parameters across 50 layers, requiring substantially more computation and memory per forward/backward pass. ResNet’s residual connections make this extra depth trainable at all, but the computational cost compared to LeNet is several orders of magnitude higher – a trade-off that is worthwhile only because ResNet targets a far more complex classification problem than LeNet was ever designed for.

21. References

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